Week 9 – Workshop
Geothermics: Transient Heat Flow

Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

Duration: 3 hours **Setting:** Workshop room

Preparation

Pre-reading:
Lecture videos:
Notes:

Purpose

This workshop develops intuition for transient heat flow: how thermal diffusivity sets the timescales and length scales over which thermal signals propagate, how half-space and plate cooling models describe the thermal evolution of oceanic lithosphere, and how latent heat modifies cooling in Stefan problems.

Learning Outcomes

Students will:

- Understand how thermal diffusivity controls timescales.
- Compare thermal and chemical diffusion processes.
- Interpret half-space and plate cooling models.
- Use Fourier series to conceptualize thermal evolution.

Session Flow (180 min)

0–45 min: Activity A: Thermal Diffusivity, Thermal Length, and Timescales

- Review $\kappa = k/(\rho c_p)$.
- Emphasize modest variation in crustal rocks.
- Contrast sediments, rock, and ice.
- Penetration depth scaling: $z \sim \sqrt{\kappa t}$.
- Borehole paleothermometry vs ice-core records.
- Depth–resolution tradeoffs.

45–75 min: Activity B: Thermal vs Chemical Diffusion

- Both obey $\sqrt{Dt}$ scaling.
- Chemical diffusion often multi-component.
- Zoning persistence vs thermal smoothing.

75–85 min: Break

85–120 min: Activity C: Half-Space Cooling

- Assumptions and boundary conditions.
- Growth of thermal boundary layer.
- Oceanic lithosphere as a thermal clock.

120–150 min: Activity D: Plate Cooling and Fourier Series

- Fixed plate thickness as boundary condition.
- Fourier series as superposition of thermal modes.
- Mode decay times and geological interpretation.

150–180 min: Activity E: Stefan Problems in Geology

- Latent heat as effective heat capacity.
- Cooling times of dikes vs sills.
- When analytic solutions break down.

Material	**Thermal Conductivity k ($\mathbf{W\,m^{-1}\,K^{-1}}$)**	**Thermal Diffusivity κ ($\mathbf{m^2\,s^{-1}}$)**	**Thermal Diffusivity κ ($\mathbf{km^2\,My^{-1}}$)**
Ice (glacial / ice sheet average)	1.6–2.0	$(0.7–0.9) \times 10^{-6}$	22–28
Clay-rich sediments	0.6–1.5	$(0.2–0.5) \times 10^{-6}$	6–16
Granite	2.5–3.5	$(0.8–1.2) \times 10^{-6}$	25–38
Basalt / Gabbro	1.7–2.2	$(0.6–1.0) \times 10^{-6}$	19–32
Peridotite (upper mantle)	3.3–4.5	$(1.0–1.5) \times 10^{-6}$	32–47

Activity A
Thermal Diffusivity, Thermal Length, and Timescales

Purpose

To understand transient heat flow in terms of:

- how far thermal signals propagate in a given time,
- how long it takes thermal changes to affect a given depth,
- and how the *thermal length scale* controls signal shape and timing.

Background

Thermal diffusivity is defined as

$$\kappa = \frac{k}{\rho c_p},$$

where k is thermal conductivity, ρ is density, and c_p is specific heat.

Values are representative ranges. Thermal conductivity and diffusivity vary with temperature, porosity, mineralogy, fluid content, and fabric. Diffusivities are calculated using $\kappa = k/(\rho c_p)$ and rounded for scaling arguments rather than precise modeling.

For diffusion-dominated heat transfer, the characteristic depth over which temperature changes propagate scales as

$$z \sim 2\sqrt{\kappa t}.$$

This thermal length sets both:

- the depth to which temperature changes have propagated after time t,
- the timescale required for a thermal perturbation at depth to influence shallower levels.

Error Function Solution

For instantaneous heating or cooling at a boundary, the temperature evolution is given by the error function solution:

$$T(z,t) = T_s + (T_b - T_s)\,\mathrm{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Important: The argument of the error function,

$$\eta = \frac{z}{2\sqrt{\kappa t}},$$

controls the *shape and timing* of the thermal signal.

Tasks

1. For timescales of 10^3, 10^5, and 10^7 years, describe how the depth of thermal influence would differ between rock and ice.
2. Explain why borehole paleothermometry in rock requires much greater depths to recover long-term climate signals than ice-core records.

Tasks

A. Distance Given Time

1. For a fixed time t, describe how temperature varies with depth z as the value of η increases.
2. Explain why temperature changes are largest for $\eta \ll 1$ and become negligible for $\eta \gg 1$.
3. Using the thermal length ℓ_T, estimate the depth to which a surface temperature change has propagated after a given time.

B. Time Given Distance

1. Rearranging the thermal length relation,
$$t \sim \frac{z^2}{\kappa},$$
explain what sets the timescale for heat to travel a distance z.
2. Consider a plume impinging on the base of the lithosphere. If the lithosphere is x km thick, what controls how long it takes for this thermal perturbation to influence surface temperatures?
3. Following delamination of a dense crustal root, heat is supplied from below. Explain why crustal heating is delayed rather than instantaneous.

C. Interpreting "Arrival" of Heat

1. Does heat arrive at a specific time, or does it diffuse gradually? Use the error function shape to justify your answer.
2. For $\eta \approx 1$, what fraction of the total temperature change has occurred? What does this imply about defining a thermal response time?
3. Discuss why the concept of a sharp thermal front is inappropriate for diffusion-dominated heat transport.

Geological Interpretation

- Thick lithosphere acts as a strong thermal filter, delaying surface response to basal heating.
- Thermal signals generated at depth are attenuated and spread in time as they propagate upward.
- The thermal length scale controls which geological processes can be observed at the surface and on what timescales.

Key Insight

In diffusion problems, time and distance are interchangeable through the thermal length $\ell_T = 2\sqrt{\kappa t}$. Heat does not "arrive" at a depth at a single moment; instead, the error function describes a gradual, scale-dependent thermal response controlled by diffusivity, distance, and boundary conditions.

Activity B
Thermal vs Chemical Diffusion

Purpose

To compare transient thermal diffusion with chemical diffusion processes familiar from igneous and metamorphic petrology.

Diffusion Framework

Both thermal and chemical diffusion can be written in the general form

$$\frac{\partial \phi}{\partial t} = D\nabla^2\phi,$$

where ϕ represents temperature or chemical concentration, and D is a diffusivity.

Tasks

1. Explain why both thermal and chemical diffusion act to smooth spatial gradients over time.
2. Chemical diffusion in minerals often preserves sharp zoning. Give two reasons why sharp compositional gradients may survive despite diffusion.
3. Describe how multi-component diffusion differs from thermal diffusion involving a single scalar field.

Comparison

- Thermal diffusion: single field, relatively uniform diffusivity.
- Chemical diffusion: multi-component, potentially rate-limited by the slowest species.

Reflection

Why is heat generally a poorer long-term recorder of geological processes than composition?

Activity C
Half-Space Cooling of the Oceanic Lithosphere

Purpose

To develop physical intuition for transient cooling of the oceanic lithosphere using the half-space cooling model.

Model Assumptions

- The mantle is initially at uniform temperature.
- Surface temperature is suddenly reduced and held fixed.
- No fixed lithospheric thickness is imposed.

Solution Form

The temperature evolution is given by

$$T(z,t) = T_m \,\mathrm{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Tasks

1. Sketch temperature–depth profiles for increasing times.
2. Identify the thermal boundary layer and describe how its thickness evolves.
3. Estimate how the depth to a given isotherm scales with seafloor age.

Geological Interpretation

- Oceanic lithosphere thickens thermally with age.
- Heat flow decreases as the plate cools away from mid-ocean ridges.

Key Insight

Explain why the half-space cooling model predicts unbounded lithospheric thickening.

Activity D
Plate Cooling and Fourier Series

Purpose

To understand how imposing a fixed lithospheric thickness modifies transient cooling and why Fourier series naturally arise in plate cooling models.

Concept

Plate models introduce a basal boundary condition at a fixed depth, representing the lithosphere–asthenosphere boundary.

Mathematical Form

The temperature field can be expressed as a Fourier series:

$$T(z,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi z}{L}\right) \exp\left(-\frac{n^2\pi^2\kappa t}{L^2}\right).$$

Tasks

1. Explain how the index n relates to thermal wavelength.
2. Why do short-wavelength modes decay faster than long-wavelength modes?
3. Compare thermal evolution predicted by plate cooling and half-space cooling.

Geological Interpretation

- Plate thickness limits lithospheric growth.
- Explains flattening of bathymetry and heat flow for old oceanic plates.

Connection

Relate Fourier modes to spectral filtering concepts used previously in gravity and magnetic field analysis.

Activity E
Stefan Problems and Cooling of Magma Bodies

Purpose

To apply transient heat flow concepts to the crystallization and cooling of magma bodies within the crust.

Physical Problem

A magma body emplaced into cooler host rock cools conductively while releasing latent heat at the solidification front.

Key Feature

The solid–liquid boundary migrates with time, making this a Stefan problem.

Tasks

1. Explain how latent heat affects the cooling rate of a magma body.
2. Compare expected cooling timescales for a narrow dike and a thick sill.
3. Identify geological situations in which analytic Stefan solutions are likely to fail.

Geological Extensions

- Magma convection.
- Episodic recharge of magma.
- Crystal-rich mush zones.

Key Insight

Geometry and diffusivity often dominate cooling timescales more strongly than radiogenic heat production.